

BONNARD, QC, Canada

WMO: 713830

Lat: 50.729N

Lon: 71.013W

Elev: 498

StdP: 95.48

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.0	-33.3	-39.9	0.1	-35.9	-36.9	0.1	-33.3	11.8	-12.7	10.5	-17.0	1.6	260	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 25.5	(d) 17.2	(e) 23.8	(f) 16.5	(g) 22.1	(h) 16.0	(i) 19.3	(j) 22.7	(k) 18.2	(l) 21.7	(m) 17.2	(n) 20.5	(o) 4.9	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	14.0	20.5	17.0	12.9	19.5	15.9	12.0	18.6	57.3	22.7	53.8	21.7	50.4	20.4	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.5	(b) 9.2	(c) 8.1	(e) -41.1	(f) 28.9	(g) 2.3	(h) 1.8	(i) -42.8	(j) 30.2	(k) -44.1	(l) 31.3	(m) -45.4	(n) 32.4	(o) -47.0	(p) 33.7		
(a) 10.5	(b) 9.2	(c) 8.1	(e) -40.8	(f) 21.1	(g) 2.4	(h) 1.7	(i) -42.5	(j) 22.3	(k) -43.9	(l) 23.3	(m) -45.2	(n) 24.3	(o) -46.9	(p) 25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-19.1	-18.3	-11.4	-2.6	5.6	12.9	15.8	14.8	10.3	3.3	-4.5	-14.0		
	DBStd	13.54	7.87	7.67	7.69	5.60	4.36	4.22	2.88	3.06	4.06	4.07	5.31	7.14		
	HDD10.0	4338	901	792	664	379	149	17	0	2	44	210	435	743		
	HDD18.3	6900	1159	1025	923	629	394	171	88	116	244	465	685	1001		
	CDD10.0	505	0	0	0	0	14	103	181	151	52	4	0	0		
	CDD18.3	27	0	0	0	0	1	7	11	7	2	0	0	0		
	CDH23.3	196	0	0	0	0	13	62	74	37	9	0	0	0		
	CDH26.7	22	0	0	0	0	2	9	9	2	1	0	0	0		
(14) Wind	WSAvg	3.5	3.4	3.5	3.6	3.6	3.3	3.5	3.3	3.4	3.8	3.9	3.8	3.3		
(15) Precipitation	PrecAvg	1006	74	38	40	60	73	109	136	109	108	77	91	85		
	PrecMax	1226	117	75	74	120	159	201	234	165	153	121	134	158		
	PrecMin	777	29	14	7	18	16	64	60	60	58	38	28	36		
	PrecStd	118	26	17	17	28	30	36	37	31	24	28	29	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	3.7	8.0	12.9	25.3	27.9	28.0	26.5	24.4	17.5	8.8	2.1		
		MCWB	3.1	2.4	3.6	5.8	15.5	18.2	19.0	18.4	18.2	13.5	7.6	1.5		
	2%	DB	0.2	0.5	3.9	9.4	20.2	25.0	25.4	24.2	21.2	13.8	6.0	0.6		
		MCWB	0.0	-0.5	1.3	4.1	12.3	16.7	17.1	17.5	16.8	11.2	5.0	0.2		
	5%	DB	-3.7	-2.3	1.8	6.9	16.6	23.0	23.6	22.2	19.0	11.8	3.7	-1.3		
		MCWB	-4.2	-3.2	0.2	3.0	10.3	15.6	16.5	16.6	15.4	9.9	2.8	-1.6		
	10%	DB	-6.8	-5.6	0.1	4.9	13.5	20.8	21.8	20.6	16.9	9.5	2.2	-3.6		
		MCWB	-7.2	-6.4	-1.1	1.9	8.8	14.4	16.0	15.9	13.8	8.0	1.6	-4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	3.0	4.6	6.7	16.7	20.6	21.0	20.2	19.6	14.0	8.3	1.7		
		MCDB	2.7	3.3	6.6	10.4	23.0	24.7	24.5	23.4	22.4	16.2	8.8	1.9		
	2%	WB	-0.3	-0.7	2.0	4.9	13.5	18.7	19.1	19.0	17.6	12.0	5.3	0.1		
		MCDB	0.0	0.4	3.3	8.3	18.5	22.8	22.3	21.9	20.4	13.5	5.7	0.4		
	5%	WB	-4.4	-3.7	0.4	3.6	11.3	17.1	17.9	17.9	16.0	10.0	3.0	-1.7		
		MCDB	-4.0	-2.7	1.6	6.2	15.7	20.8	21.6	20.9	18.1	11.3	3.4	-1.4		
	10%	WB	-7.4	-6.5	-1.2	2.4	9.3	15.4	17.0	16.8	14.3	8.1	1.6	-4.0		
		MCDB	-7.0	-5.8	0.1	4.5	12.9	19.2	20.5	19.5	16.4	9.3	2.1	-3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	13.6	14.1	11.6	11.1	11.0	9.9	9.2	8.6	6.1	6.0	9.0		
		MCDBR	11.2	12.2	13.9	13.8	17.3	14.3	12.7	12.0	11.4	9.9	6.8	8.2		
		MCWBR	10.7	11.5	11.5	9.4	10.3	7.2	6.3	6.4	7.1	7.4	6.2	8.3		
		MCDBR	10.7	11.9	11.6	11.5	15.6	11.4	10.2	9.6	10.0	8.5	6.7	8.3		
	5% WB	MCWBR	10.7	11.0	9.9	8.4	10.2	6.8	5.9	6.0	7.2	6.9	6.4	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.281	0.326	0.340	0.368	0.378	0.360	0.340	0.309	0.283	0.241		
	taud		2.266	2.267	2.238	2.231	2.432	2.403	2.390	2.453	2.487	2.543	2.416	2.302		
	Ebn at Noon		799	893	925	914	916	888	872	873	854	825	749	749		
	Edh at Noon		66	87	110	125	108	112	112	100	87	68	59	55		
(44) All-Sky Solar Radiation	RadAvg		1.20	2.31	3.68	4.80	5.39	5.42	4.91	3.97	2.99	1.70	0.87	0.70		
	RadStd		0.14	0.20	0.32	0.36	0.39	0.44	0.43	0.25	0.23	0.17	0.10	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page